

Strong magnetic fields and contact-interacting few-fermion systems

S. Mondal, M. Elyahu, N. Barnea, and J. Kirscher

My aim is to predict the critical number of particles which form a self-bound state (droplet) while their pair interaction is not attractive enough to sustain a bound dimer. As an intermediate step, I analyze the critical background-field strength for the in-vacuum unstable A -body system to become bound. This field strength shall be analyzed for Bose and Fermi particles.

SCRATCH PAD

$$\mathbf{A} = \frac{1}{2} \mathbf{B} \times \mathbf{r} \tag{1}$$

$$= \frac{1}{2} B \rho \hat{\phi} \tag{for } \mathbf{B} = B \hat{z}$$

$$\text{with } \rho = \sqrt{x^2 + y^2} \tag{2}$$

I. INTRODUCTION – WHAT FASCINATES US?

Assume a collection of A non-interacting, non-relativistic identical bosons and imagine you could tune-on a spherically symmetric, attractive pair interaction smoothly. The regime of large enough interaction strength ν to form a dimer with zero energy and the associated universal 3-body (original: [1, 2]; alternative, potentially generalizable explanation: [3, 4]) and cluster spectra (empirical/numerical: [5–10]; field-theoretical and classical approach: [11, 12])

II. THEORETICAL DESCRIPTION OF STRONGLY INTERACTING FERMIONS IN A MAGNETIC BACKGROUND FIELD

In order to describe the effect of a static magnetic field, which is not necessarily perturbatively small, on the low-energy spectra of small nuclei, the effective-field-theory Hamiltonian of the latter is augmented by, first, minimally coupling the derivatives, and second, by the magnetic-moment 1-body interaction (a remnant of the non-relativistic reduction of the Dirac equation) and the coupling of the $\mathbf{B}$ field to the four-nucleon leading-order momentum-independent vertex.

$$\hat{H} = \tag{3}$$

The so-called Landau spectrum of the minimally coupled, non-interacting limit of this Hamiltonian can be derived analytically

$$E = \tag{4}$$

A. Notable features

1. Total Spin is not conserved, which means that if we consider a 2-nucleon system, it resides in a superposition of spin-1 and spin-0 states.
2. Bound states might not have negative eigenvalues because the non-interacting theory still contains the effect of the magnetic field on the charged particles and shifts their ground state by, at least, the energy in the lowest Landau level.
- 3.

III. TWO-NUCLEON SYSTEMS

First, we analyse the predictions in Ref. [13], *i.e.*, what is the dependence of the lowest eigenvalues of the 2-proton system on the strength of the magnetic field?

IV. THREE NUCLEONS

How does the critical field strength at which the 3-proton system becomes bound compare with the corresponding strength which is necessary to bind two protons?

V. FOUR NUCLEONS

Same critical-strength ratio? Now we extrapolate and predict that no more than $|\mathbf{B}| \sim 4$ T is needed to bind an A -proton cluster.

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- [1] V. Efimov, Phys. Lett. B **33**, 563 (1970).
 - [2] V. Efimov, Nucl. Phys. **A210**, 157 (1973).
 - [3] R. Amado and J. Noble, Physics Letters B **35**, 25 (1971).
 - [4] S. K. Adhikari and R. D. Amado, Phys. Lett. B **40**, 11 (1972).
 - [5] J. von Stecher, Journal of Physics B: Atomic, Molecular and Optical Physics **43**, 101002 (2010).
 - [6] J. von Stecher, Phys. Rev. Lett. **107**, 200402 (2011).
 - [7] J. Horne, J. A. Salas, and K. Varga, Few-Body Systems **55**, 1245 (2014).
 - [8] M. Gattobigio and A. Kievsky, Phys. Rev. A **90**, 012502 (2014).
 - [9] Y. Yan and D. Blume, Phys. Rev. A **92**, 033626 (2015).
 - [10] M. Yamashita, D. Fedorov, and A. Jensen, Physical Review A **81** (2009), 10.1103/PhysRevA.81.063607.
 - [11] D. T. Son, M. Stephanov, and H.-U. Yee, Phys. Rev. A **106**, L050801 (2022), arXiv:2112.03318 [nucl-th].
 - [12] K. Wang, T. Preis, and D. T. Son, (2024), arXiv:2407.03102 [cond-mat.quant-gas].
 - [13] W. Detmold, K. Orginos, A. Parreño, M. J. Savage, B. C. Tiburzi, S. R. Beane, and E. Chang (NPLQCD Collaboration), Phys. Rev. Lett. **116**, 112301 (2016).